

Vulvovaginal and Cervical Histology and Pathology

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Histology Resources

Michigan Histology Laboratory Manual [A link to an online histology lab guide](#)

Histology Guide <https://histopathology.pennstatehealth.net/slidebox/slidebox.html>

Digital Histology <https://digitalhistology.org/>

Histology Primer(American Board of Pathology) <https://histologyprimer.abpath.org/Modules/Index>

Shotgun Histology [A link to a series of Youtube videos](#)

University of Utah <https://webpath.med.utah.edu/HISTHTML/HISTO.html>

Duke University Histology <https://histology.oit.duke.edu/>

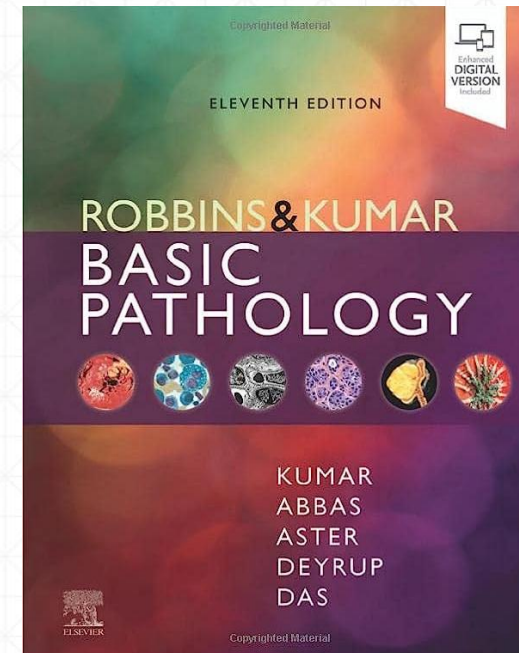
University of Leeds UK <https://www.histology.leeds.ac.uk/index.php>

Netter's Histology Textbook

Wheater's Histology Textbook

Pathology Resources

- PathElective <https://www.pathelective.com/>
- Kurt's Notes <https://schaberg.faculty.ucdavis.edu/>
- Interactive Pathology Lab <https://peir.path.uab.edu/wiki/IPLab>
- **WebPathology.com** <https://www.webpathology.com/index.asp#>
- **Robbins Pathology Textbook (11 th. Edition.)**



Objectives:

- Correlate the histological structure of the cervix with its function.
- Predict functional outcomes that result from altered structure and function of the uterine cervix (e.g., cervical cancer).
- Correlate the histological structure of the vagina with its functions.
- Correlate the histological structure of greater vestibular (Bartholin's) glands and paraurethral glands with their functions.
- Discuss common pelvic infections, including those affecting the vulva, vagina, and cervix, and describe the pathogenesis of these diseases, common organisms involved, and related complications.
- List the differential diagnosis for benign and neoplastic lesions of the vulvar epithelium as well as the clinical and pathologic features of each, including lichen sclerosus, squamous cell hyperplasia, vulvar intraepithelial neoplasia, vulvar carcinoma, and extramammary Paget disease.
- Describe the clinicopathologic features of vaginal embryonal rhabdomyosarcoma (sarcoma botryoides).
- Discuss the common human papillomavirus types that affect the cervix and the pathogenesis of cervical dysplasia, neoplasia, cervical screening methods, and prevention.
- Describe the morphologic features of the spectrum of cervical dysplasia and neoplasia derived from both cytologic and tissue specimens and clinical outcomes associated with each.

Question 1:

A 23-year-old woman presents with mucopurulent vaginal discharge and post-coital bleeding. On pelvic exam, her cervix appears erythematous with scattered petechiae. NAAT confirms the presence of a protozoan.

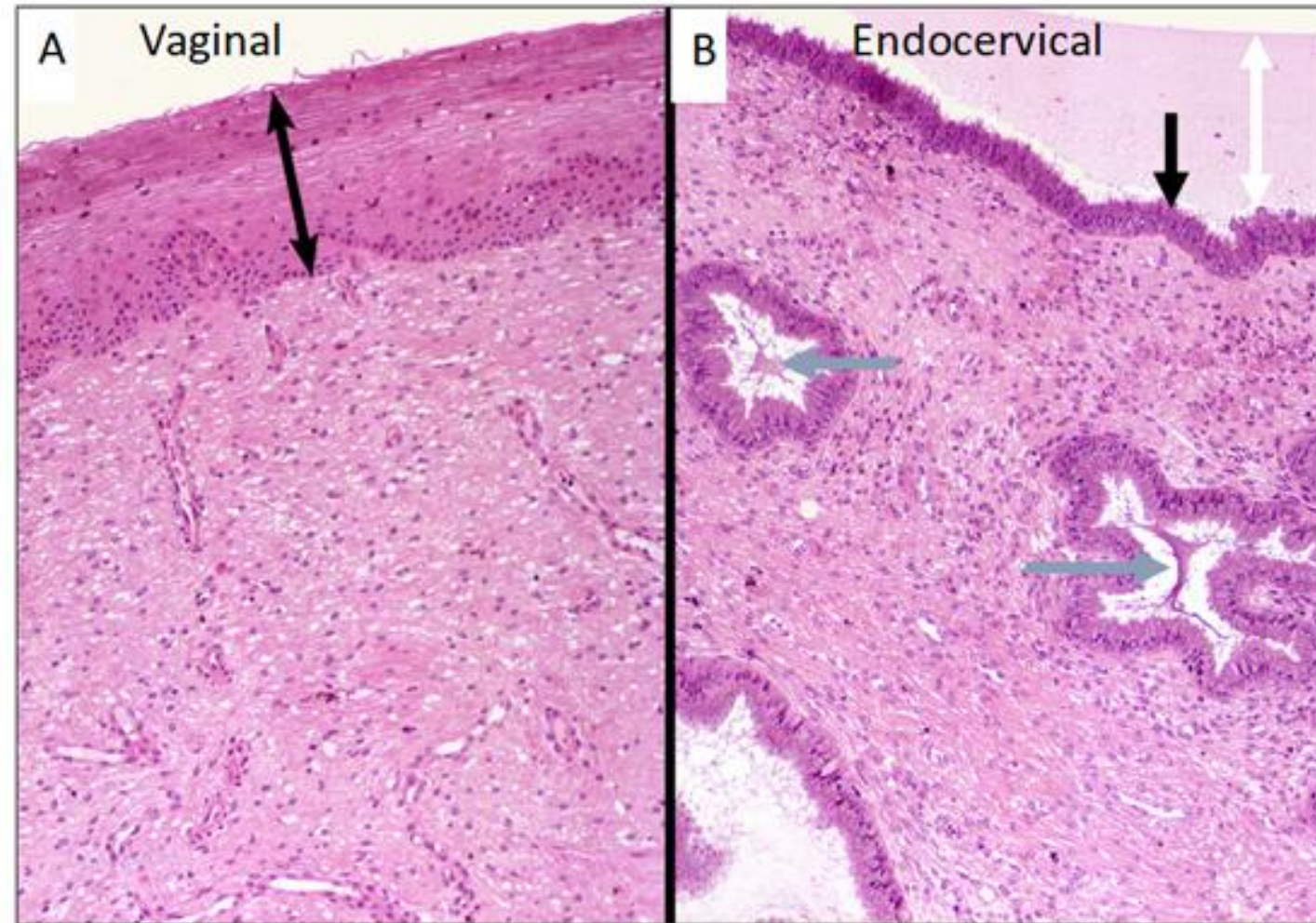
Which organism is the most likely cause of her symptoms?

- A. Chlamydia trachomatis
- B. Neisseria gonorrhoeae
- C. Trichomonas vaginalis
- D. Herpes simplex virus type 2
- E. Ureaplasma urealyticum

Functional Histology of the Cervix

LO: Correlate the histological structure of the cervix with its function.

- Cervix is lined by columnar epithelium (endocervix) + squamous epithelium (ectocervix).
- Provides a barrier and mucus secretion to regulate sperm entry and protect the upper genital tract.
- The ectocervix, like the vagina (i.e., the part of the cervix protruding into the vagina), is lined with squamous epithelium
- The epithelial lining of the endocervical canal is columnar mucous-secreting glandular epithelium
- This epithelium infolds to form the plicae palmatae ("palm folds"), also known as cervical crypts or glands.
- In the presence of high estrogen levels at ovulation, the cells lining the crypts secrete abundant alkaline secretions (grey arrows). This mucus lubricates the vagina and nourishes spermatozoa.
- Inflammation alters epithelial integrity → friability, bleeding, ↑ susceptibility to ascending infection, and progression to neoplasia (esp. HPV-related).



Cervicitis: Definition, Types, Etiology

LO: Discuss common pelvic infections, including those affecting the vulva, vagina, and cervix, and describe the pathogenesis of these diseases, common organisms involved, and related complications.

Definition & Types:

- Inflammation of the ectocervix. Acute = mostly infectious. Chronic = often noninfectious (chemical, mechanical, autoimmune).

Infectious (most common):

- Chlamydia trachomatis (most common, ~40%).
- Neisseria gonorrhoeae.
- Trichomonas vaginalis → classic “strawberry cervix” (punctate hemorrhages).
- HSV-2 → vesicles/ulcers.
- HPV (some types), Ureaplasma urealyticum.
- Associated with bacterial vaginosis.

Noninfectious causes:

- **Chemical:** soaps, detergents, spermicides, douches, contraceptive creams.
- **Mechanical:** condoms, diaphragms, caps, tampons.
- **Systemic inflammatory diseases:** lichen planus, Behçet syndrome.

Cervicitis: C/P, Diagnosis, Treatment

Clinical Presentation:

- asymptomatic.
- Symptoms: mucopurulent discharge (leukorrhea), post-coital bleeding, dyspareunia, pelvic/abdominal pain.

Exam findings:

- Yellow mucoid discharge from os.
- Swollen cervix with red punctate hemorrhages → “Strawberry cervix” (especially *Trichomonas*).
- HSV → vesicles/ulcers.

Complications:

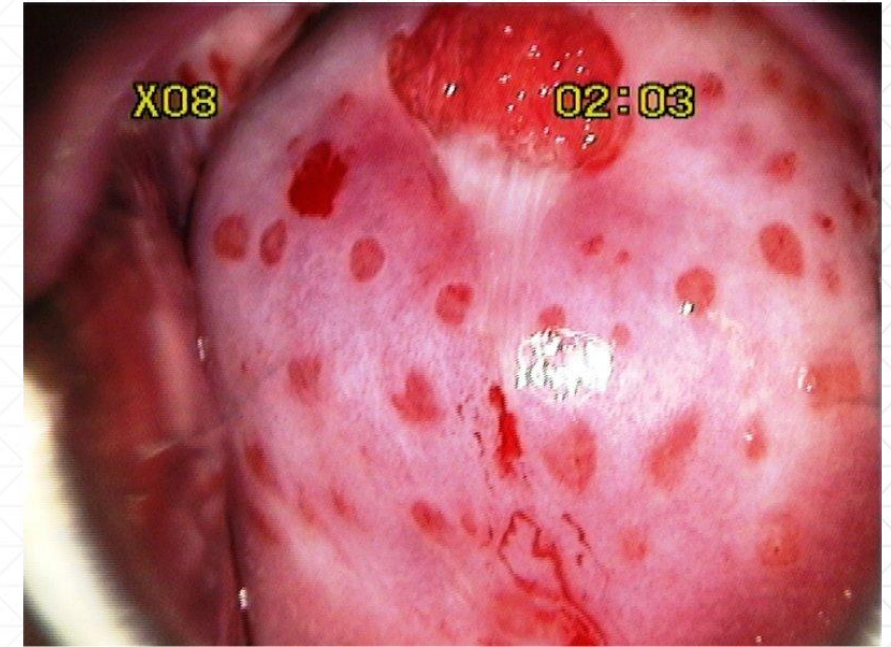
- linked with pelvic inflammatory disease (PID), infertility, ↑ HIV transmission risk.

Diagnosis

- **Nucleic Acid Amplification Test (NAAT)** = gold standard (detects Chlamydia, Gonorrhea, *Trichomonas*).
- Samples: vaginal secretions or urine.
- Important to rule out co-infections with BV, Candida, or normal flora.

Treatment

- Empiric antibiotics (covering chlamydia + gonorrhea) for high-risk patients or when pathogen not detected.
- Specific therapy based on pathogen once identified.
- Partner treatment (esp. trichomoniasis, gonorrhea, chlamydia).



Infection with *Trichomonas vaginalis* may produce a “strawberry” appearance of the cervix because of focal round patches of dilated capillaries on the surface.

Question 2:

A 62-year-old woman with urinary incontinence develops intense vulvar pruritus after using antiseptic wipes. The exam reveals well-defined erythematous, weeping, and crusting papules/plaques on the vulva. No fever.

What is the most likely cause?

- A. Allergic/contact irritant dermatitis
- B. *Candida* vulvitis
- C. HSV-2 infection
- D. *Neisseria gonorrhoeae* infection of vulvovaginal glands
- E. HPV 16 infection

Question 3:

A 29-year-old woman presents with vaginal pruritus and thin, homogeneous discharge with a fishy odor. Vaginal pH is elevated (>4.5), and microscopy reveals clue cells.

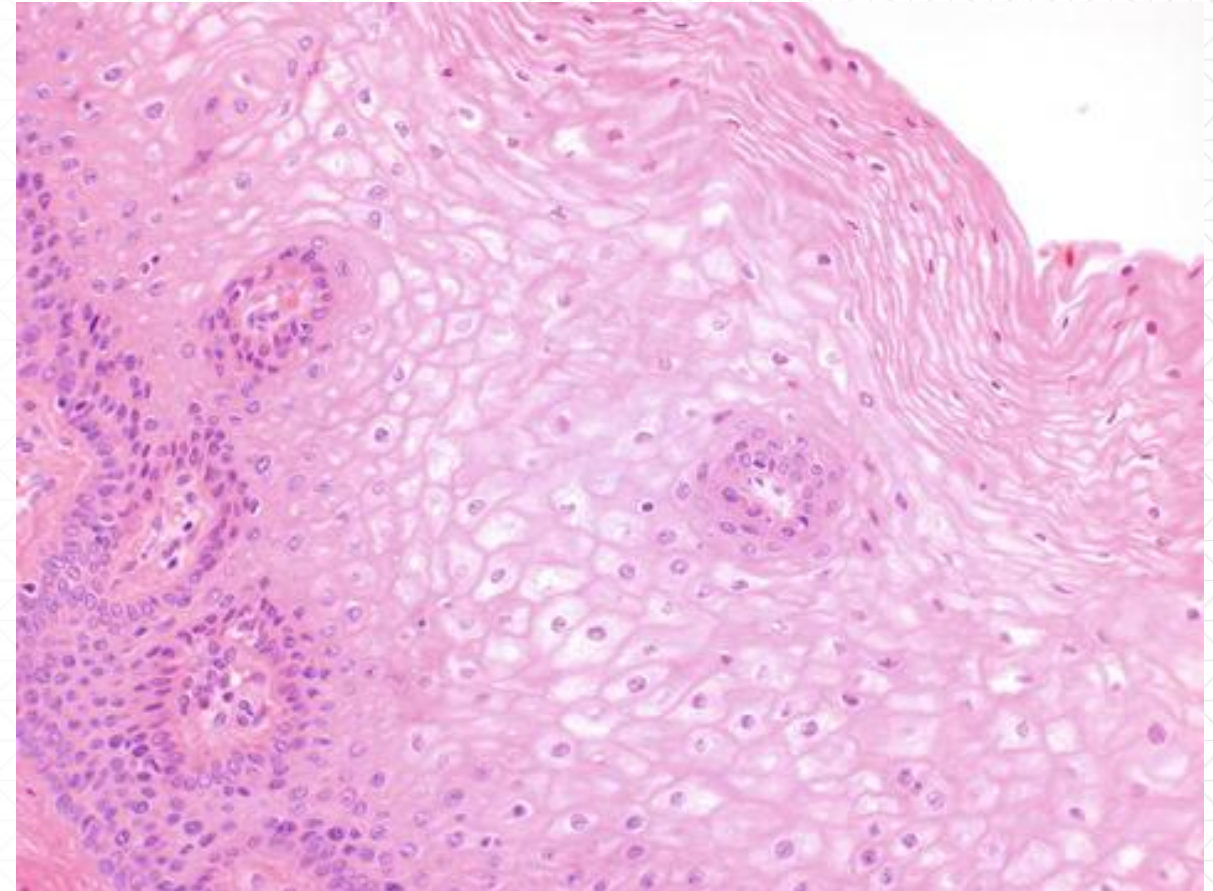
Which of the following best explains the pathogenesis of this condition?

- A. Overgrowth of anaerobic bacteria disrupting normal lactobacilli flora
- B. Invasion of vaginal epithelium by *Candida albicans* hyphae
- C. Viral cytopathic effect leading to ulceration of mucosal surfaces
- D. Protozoal motility causing epithelial disruption and inflammation
- E. Estrogen deficiency leading to thinning of vaginal epithelium

Histological Structure–Function Correlation of the Vagina:

LO: Correlate the histological structure of the vagina with its functions.

- Stratified squamous epithelium (non-keratinized): protects against trauma & infection.
- Glycogen-rich (during reproductive life) epithelial cells: metabolized by Lactobacilli → lactic acid → acidic pH (protective barrier).
- Absence of glands: vaginal lubrication derived from cervical and vestibular glands.
- Mucosal immunity: supports balance of microbiota and defense against pathogens.



Vaginitis: Etiology, Pathogenesis, Pathophysiology

LO: Discuss common pelvic infections, including those affecting the vulva, vagina, and cervix, and describe the pathogenesis of these diseases, common organisms involved, and related complications.

Etiology:

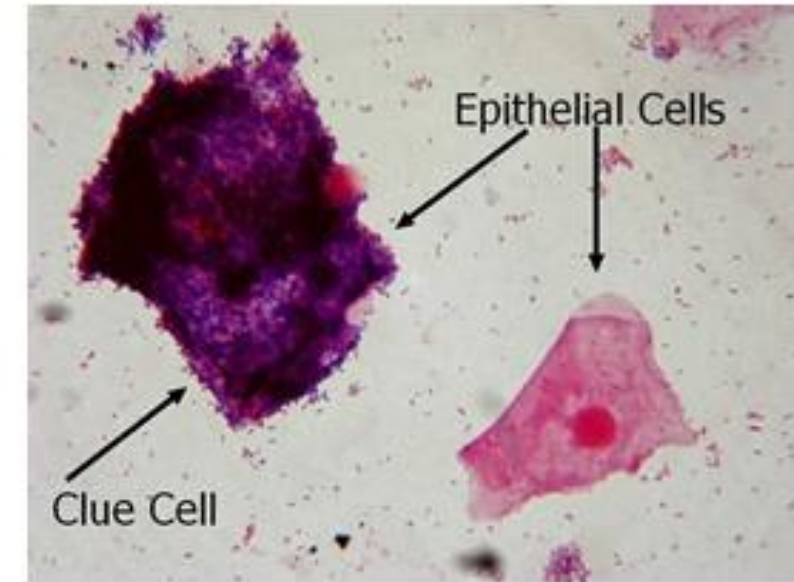
- Infectious causes: Bacterial vaginosis (BV): overgrowth of *Gardnerella vaginalis* & anaerobes after ↓ *Lactobacillus*.
- Vulvovaginal candidiasis (VVC): *Candida albicans* overgrowth triggered by antibiotics, hormones, immunosuppression.
- Trichomoniasis: *Trichomonas vaginalis* (protozoan STI).
- Noninfectious causes: atrophic vaginitis (low estrogen), allergic/chemical irritation, foreign body, desquamative inflammatory vaginitis, lichen planus.

Pathogenesis:

- BV: Loss of protective H₂O₂-producing *Lactobacillus* → ↑ anaerobes (e.g., *Gardnerella*).
- VVC: *Candida* shifts from commensal yeast → pathogenic hyphae.
- Trichomoniasis: direct protozoal infection.
- Common mechanism: oxidative stress → ROS, cytokine release (TNF-α, IL-1β, IL-6) → epithelial injury & inflammation.

Pathophysiology & Clinical Features:

- BV: Thin, gray-white, fishy discharge; ↑ pH > 4.5; clue cells on microscopy.
- VVC: Thick, “cottage-cheese” discharge; erythema; yeast & pseudohyphae on KOH prep.
- Trichomoniasis: Frothy yellow-green discharge, strawberry cervix, motile flagellates on wet mount.
- Atrophic vaginitis: Postmenopausal; dryness, burning, dyspareunia; thin mucosa, loss of rugae, ↑ pH.



Two epithelial cells are seen; the one on the left is 'coated' with gram variable bacilli (*Gardnerella vaginalis*) making the cell appear purplish. These are usually seen with some frequency in a patient experiencing bacterial vaginosis.

Vaginitis: Complications, Diagnosis, Treatment

LO: Discuss common pelvic infections, including those affecting the vulva, vagina, and cervix, and describe the pathogenesis of these diseases, common organisms involved, and related complications.

Complications:

- ↑ Risk of pelvic inflammatory disease (PID), adverse pregnancy outcomes (preterm labor, low birth weight).
- Trichomoniasis increases susceptibility to HIV and other STIs.
- Chronic/recurrent infections affect quality of life.

Diagnosis

- BV: Amsel's criteria (discharge, pH >4.5, clue cells, whiff test), Nugent score on Gram stain.
- VVC: KOH prep (budding yeast, pseudohyphae); culture for non-albicans species.
- Trichomoniasis: Wet mount (motile protozoa), NAAT.
- Atrophic: Clinical exam, microscopy (↑ WBCs, parabasal cells, ↓ Lactobacilli).

Treatment

- BV: Metronidazole or clindamycin (oral or topical).
- VVC: Oral fluconazole (single dose) or topical azoles; longer regimens for complicated/recurrent cases.
- Trichomoniasis: Metronidazole/tinidazole (single dose); treat partners.
- Atrophic vaginitis: Topical vaginal estrogen; lubricants/moisturizers.

Question 4:

A 27-year-old woman presents with severe unilateral vulvar pain and swelling. The exam reveals a tender, fluctuant mass at the posterolateral vestibule near the labia minora. You explain that obstruction of the affected gland's duct can lead to abscess formation.

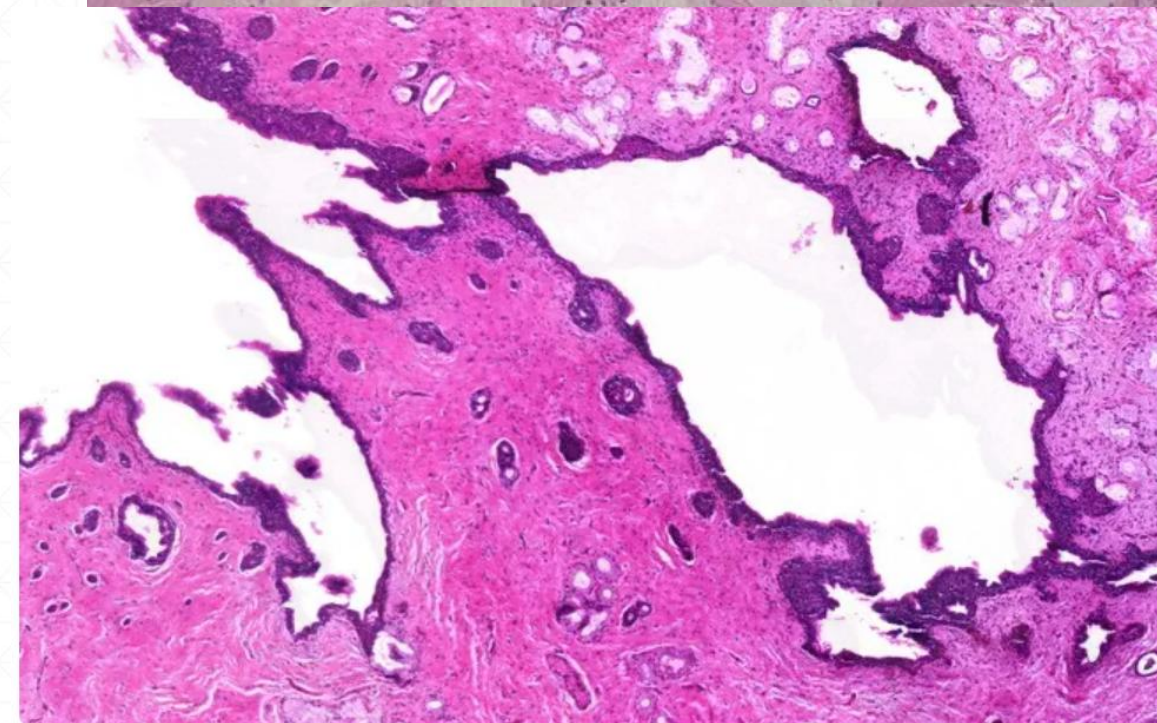
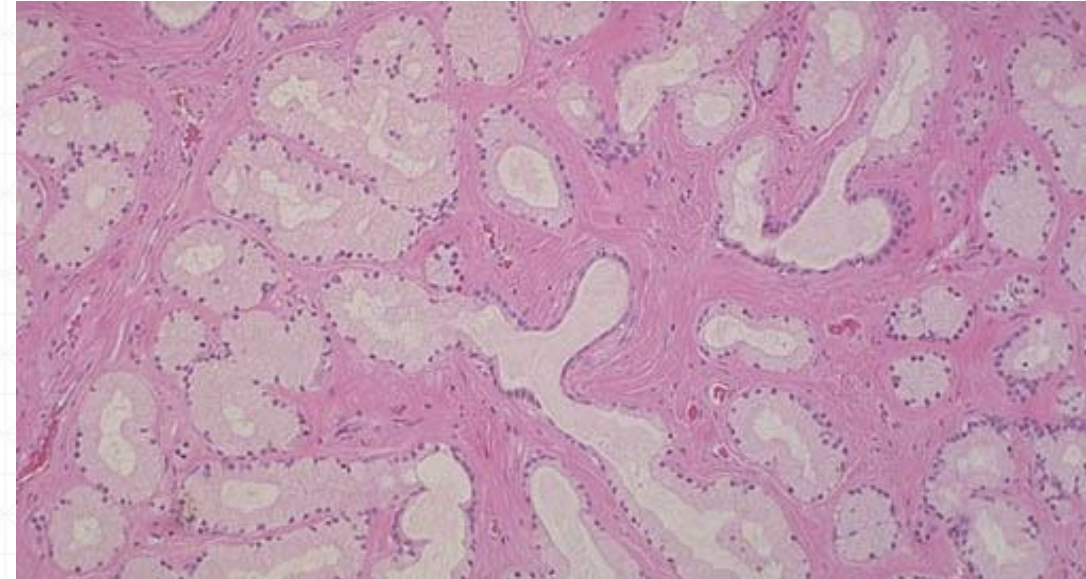
Which of the following best correlates the histologic structure and function of the affected gland ?

- A. Mucus-secreting tubuloalveolar glands that lubricate the vaginal opening; similar to smaller glands near the urethra that help moisten its orifice
- B. Serous acini that secrete antimicrobial peptides into the vaginal canal; unlike nearby sebaceous glands that produce lipids
- C. Apocrine glands involved in scent signaling; unlike adjacent glands that secrete enzymes into the bladder
- D. Squamous-lined crypts that produce keratin; unlike neighboring ciliated ducts that propel secretions
- E. Endocrine cords that release hormones into the lamina propria; unlike surrounding glands that regulate pH

Bartholin's Glands:

LO: Correlate the histological structure of greater vestibular (Bartholin's) glands and paraurethral glands with their functions.

- Location of the swelling: The mass is described as being at the posterolateral vestibule near the labia minora.
- This is the classic location of the Bartholin glands, which are situated at the 5 and 7 o'clock positions of the vaginal introitus.
- The **mucinous glands** seen here make up the Bartholin's gland in the wall of the vagina.
- This small gland is about 1 cm in size and the secretions drain via a **duct** lined by cuboidal to transitional epithelium.
- Function and duct obstruction: Bartholin glands secrete mucus for vaginal lubrication, especially during sexual arousal.
- Their ducts open into the vestibule, and **duct obstruction can lead to cyst formation (transitional epithelium) or abscess**, which presents as painful, fluctuant swelling.
- Clinical presentation: Unilateral pain and swelling with a fluctuant mass is typical of a Bartholin gland abscess.
- Paraurethral gland (Skene) glands: infections are less common and usually present with urethral symptoms (e.g., dysuria, purulent discharge from the urethra), not posterolateral vestibular swelling.



Question 5:

A 68-year-old woman presents with chronic vulvar itching and pain during intercourse. On exam, there are white, thin plaques with areas of skin fragility and scarring. Biopsy reveals thinning of the epidermis, loss of rete ridges, homogenization of the dermal collagen, and a band-like lymphocytic infiltrate.

Which of the following is the most likely diagnosis?

- A. Lichen sclerosus
- B. Squamous cell hyperplasia
- C. Vulvar intraepithelial neoplasia
- D. Vulvar carcinoma
- E. Extramammary Paget disease

Question 6:

A 45-year-old woman presents with thickened, leathery vulvar skin and intense pruritus. Biopsy shows hyperkeratosis, acanthosis, and elongation of rete ridges.

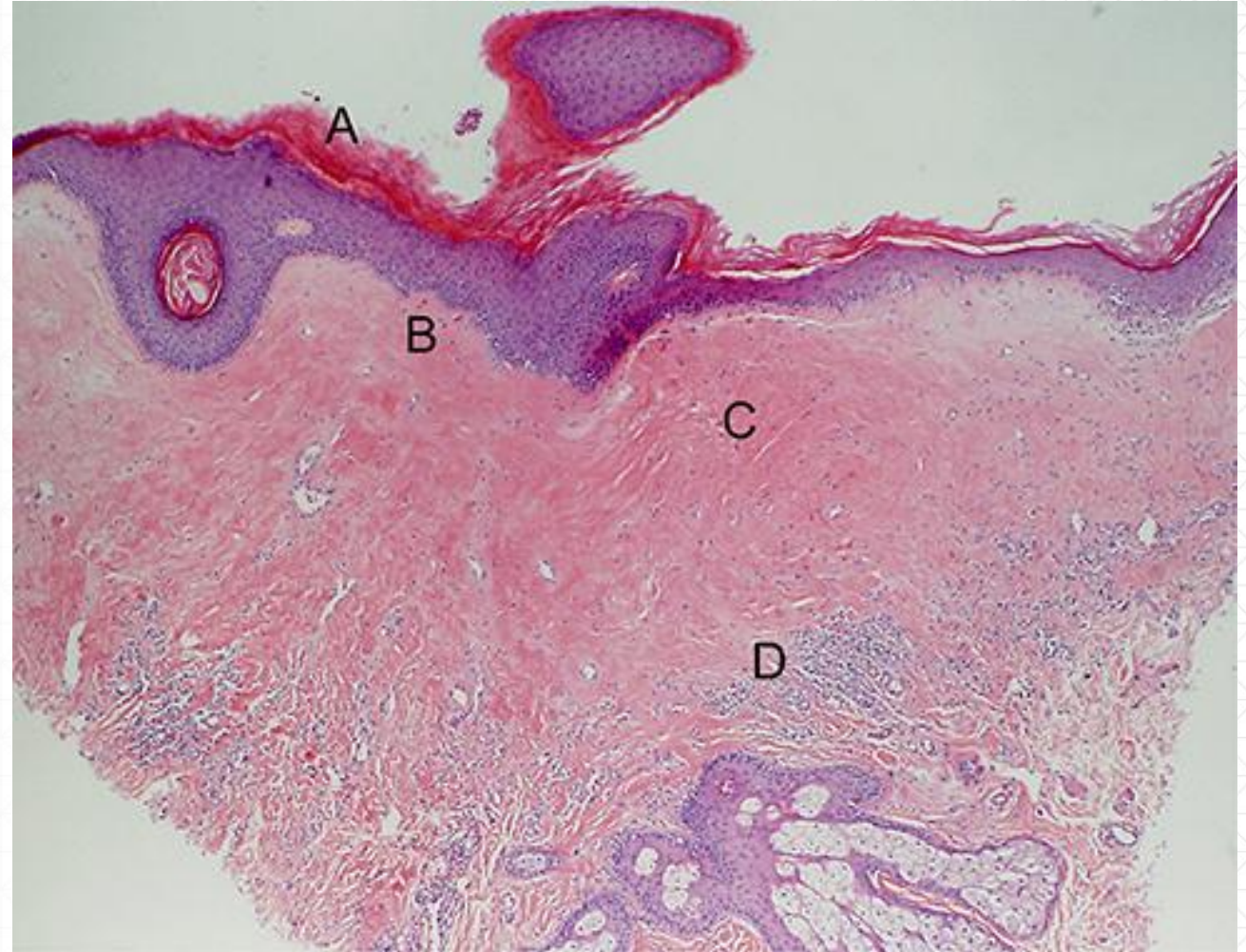
Which condition is most consistent with these findings?

- A. Lichen sclerosus
- B. Squamous cell hyperplasia
- C. Vulvar carcinoma
- D. Paget disease
- E. HSV infection

1. Lichen Sclerosus:

LO: List the differential diagnosis for benign and neoplastic lesions of the vulvar epithelium as well as the clinical and pathologic features of each, including lichen sclerosis, squamous cell hyperplasia, vulvar intraepithelial neoplasia, vulvar carcinoma, and extramammary Paget disease.

- Smooth, white plaques (leukoplakia), atrophy, labial stiffening, narrowing of vaginal orifice.
- Pathology: Epidermal thinning, loss of rete ridges, dermal fibrosis, bandlike lymphocytic infiltrate.
- Common in postmenopausal women & prepubertal girls.
- Autoimmune association.
- Risk: Slight ↑ risk of squamous cell carcinoma, but not premalignant itself.



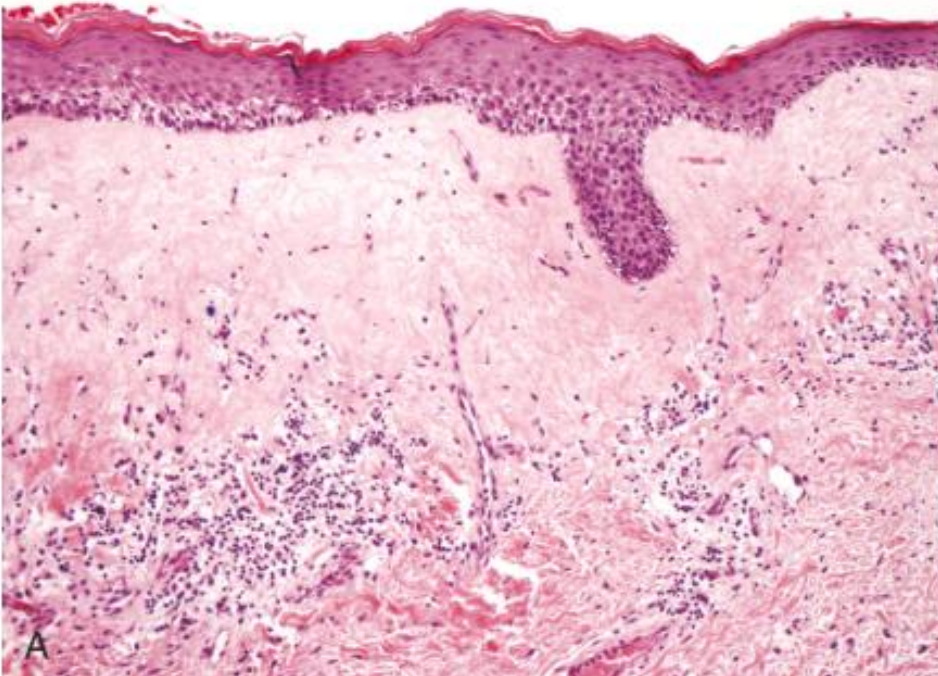
Hyperkeratosis of the epidermis (A), epidermal atrophy with loss of rete ridges (B), homogenization of the collagen below the dermal-epidermal junction (C), and lichenoid T-lymphocyte infiltrate (D).

2. Squamous Cell Hyperplasia (Lichen Simplex Chronicus)

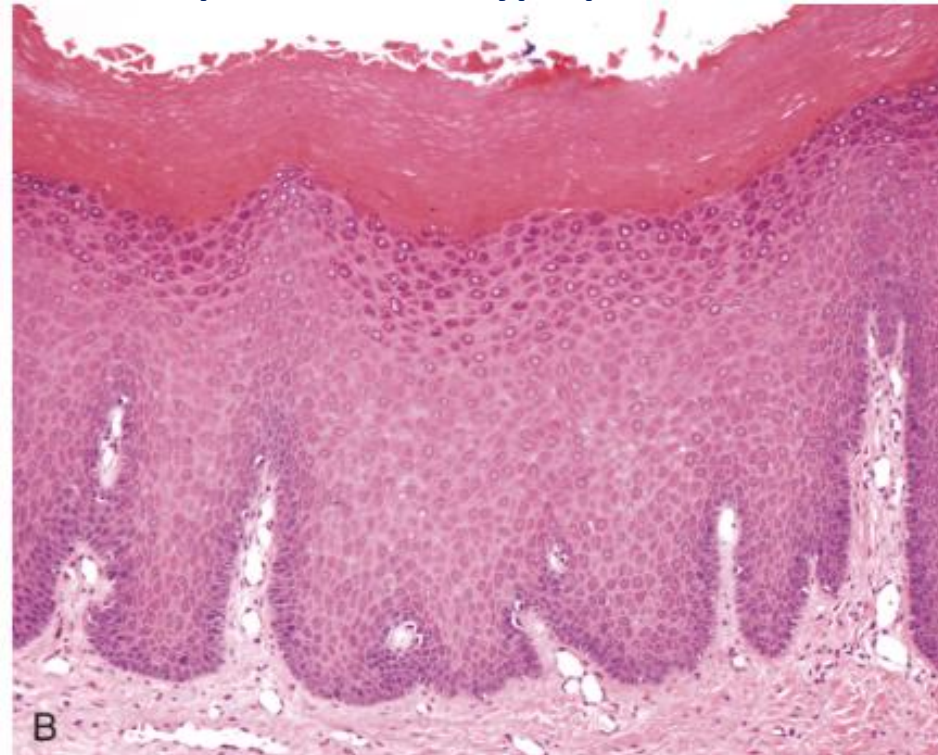
LO: List the differential diagnosis for benign and neoplastic lesions of the vulvar epithelium as well as the clinical and pathologic features of each, including lichen sclerosus, squamous cell hyperplasia, vulvar intraepithelial neoplasia, vulvar carcinoma, and extramammary Paget disease.

- Leukoplakia due to chronic irritation/pruritus.
- Pathology: Epidermal thickening (acanthosis), elongated rete ridges, hyperkeratosis.
- Notes: Nonspecific; no cytologic atypia.
- Must biopsy to exclude VIN or carcinoma.

Lichen Sclerosus



Squamous Cell Hyperplasia



Question 7:

A 21-year-old woman has multiple exophytic, verrucous papules on the labia minora. Biopsy shows papillary exophytic stroma covered by thickened squamous epithelium with koilocytosis.

Which statement is most accurate?

- A. Caused by high-risk HPV 16/18 with frequent malignant progression
- B. Often multifocal; lesions commonly regress without therapy
- C. Caused by low-risk HPV 6/11 and do not progress to cancer
- D. Associated with basal atypia and TP53 mutation (dVIN)
- E. Best prevented only by bivalent vaccine

Question 8:

A 73-year-old woman with long-standing lichen sclerosis has a solitary leukoplakic vulvar plaque. Biopsy shows basal atypia with parakeratosis and irregular rete ridges; tumor shows aberrant p53 and absent/focal p16.

Which is the most likely pathway?

- A. HPV-associated HSIL → warty/basaloid SCC
- B. HPV-independent dVIN → keratinizing SCC
- C. Paget disease with CK7+/GATA3+ cells
- D. Condyloma acuminatum with Ki-67 in upper third
- E. HPV 6/11–associated VIN

Question 9:

A 68-year-old woman presents with a persistent red, scaly vulvar plaque that has not responded to topical corticosteroids. Biopsy reveals intraepidermal atypical cells. Immunostaining is positive for CK7 and CEA, and negative for S100 and HMB-45.

Which of the following is the most likely diagnosis?

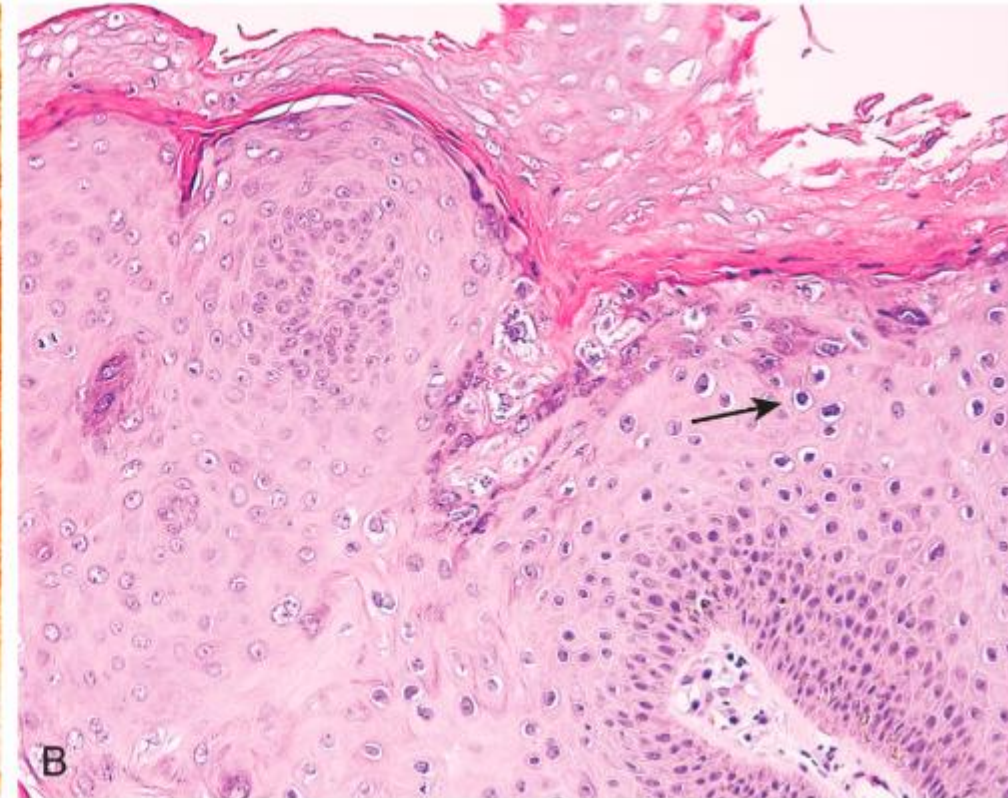
- A. Vulvar melanoma
- B. Squamous cell carcinoma
- C. Extramammary Paget disease
- D. Vulvar intraepithelial neoplasia (VIN)
- E. Chronic contact dermatitis

3. Vulvar Condylomas (HPV-related)

LO: List the differential diagnosis for benign and neoplastic lesions of the vulvar epithelium as well as the clinical and pathologic features of each, including lichen sclerosus, squamous cell hyperplasia, vulvar intraepithelial neoplasia, vulvar carcinoma, and extramammary Paget disease.

- Warty, exophytic papules; red-pink to hyperpigmented.
- Etiology: Condyloma lata → syphilis (rare).
- Condyloma acuminata → low-risk HPV (6, 11).
- Pathology: Papillary squamous proliferation with koilocytosis (perinuclear halos, wrinkled nuclei).
- Notes: Benign, do not progress to cancer, but signal HPV exposure.

A) Numerous condylomas of the vulva. (B) Histopathologic features of condyloma acuminatum include acanthosis, hyperkeratosis, and human papillomavirus cytopathic effect (koilocytic atypia) characterized by atypical, enlarged, hyperchromatic nuclei with perinuclear halos (arrow).



VIN and Vulvar SCC

LO: List the differential diagnosis for benign and neoplastic lesions of the vulvar epithelium as well as the clinical and pathologic features of each, including lichen sclerosus, squamous cell hyperplasia, vulvar intraepithelial neoplasia, vulvar carcinoma, and extramammary Paget disease.

4. Vulvar Intraepithelial Neoplasia (VIN)

HPV-related (classic VIN):

Younger women (~60), high-risk HPV (16).

Precancerous lesion → basaloid/warty carcinoma.

HPV-negative (differentiated VIN, dVIN):

Older women (~75), associated with lichen sclerosus or squamous hyperplasia.

Precursor to keratinizing SCC.

5. Vulvar Squamous Cell Carcinoma

Epidemiology: ~3% of female genital cancers; >60 yrs.

Types:

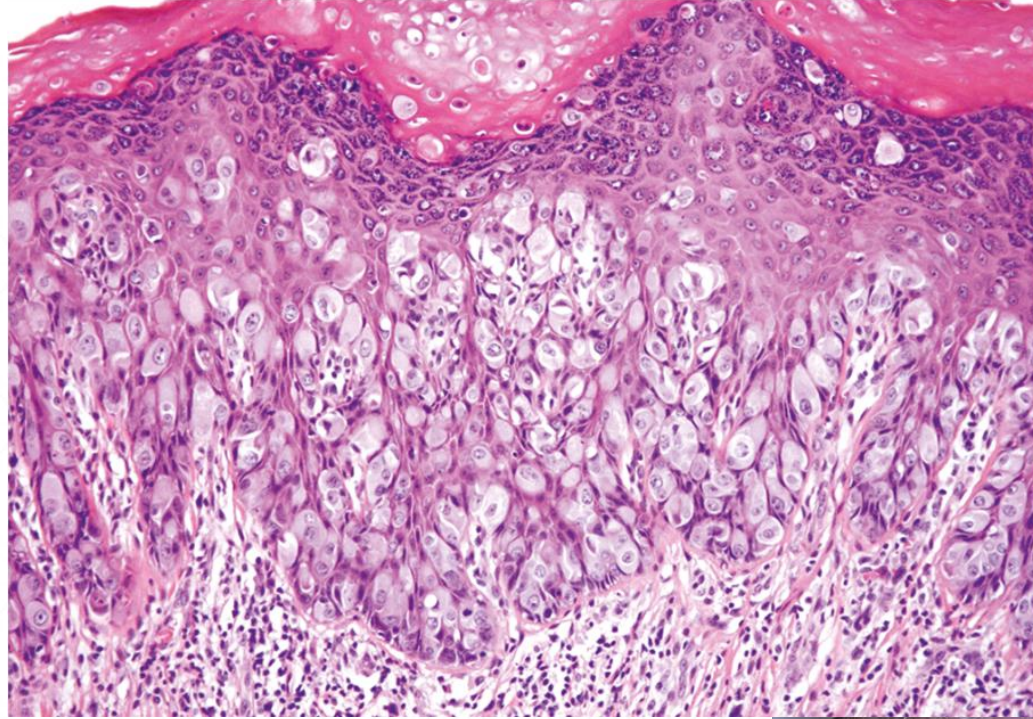
HPV-positive: basaloid/warty, multifocal, poorly differentiated.

HPV-negative: keratinizing, unifocal, well-differentiated, arises in background of lichen sclerosus/dVIN.

Spread: Local invasion → lymph nodes → distant metastasis.

5. Extramammary Paget Disease

- Clinical: Red, scaly, eczematoid plaque; mimics dermatitis.
- Pathology: Large pale-staining malignant epithelial (Paget) cells within epidermis, singly and in clusters.
- Notes: Rare intraepithelial adenocarcinoma of vulva; usually not associated with underlying carcinoma (unlike breast Paget).
- Course: Indolent but may persist for years; invasive cases carry poor prognosis.



Large tumor cells with pale-pink cytoplasm are seen infiltrating the epidermis. Chronic inflammatory cells are present in the underlying dermis.



Question 10:

A 3-year-old girl presents with vaginal bleeding and a grape-like mass protruding from the vagina. Histology shows small round blue cells in a cambium layer beneath the epithelium.

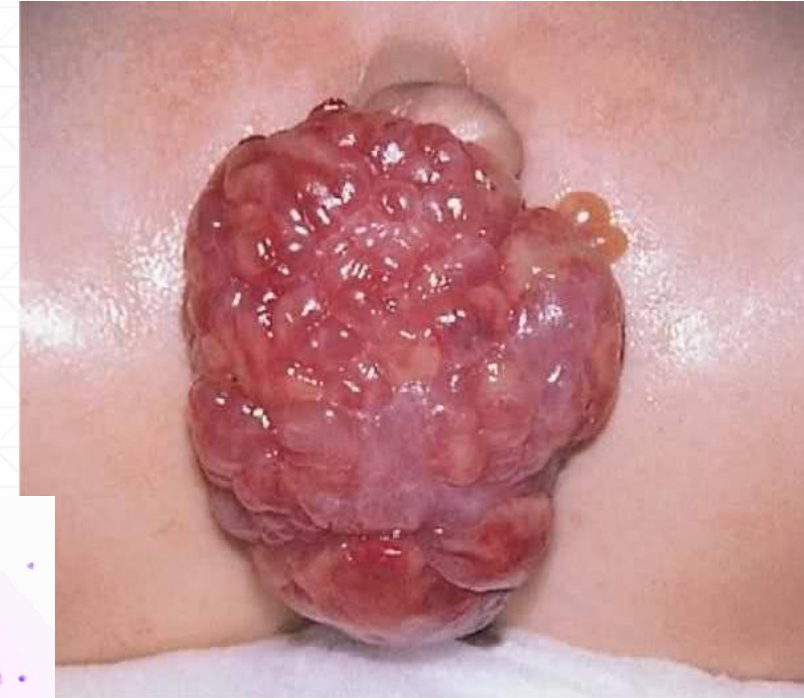
Which diagnosis is most likely?

- A. Clear cell carcinoma
- B. Endodermal sinus tumor
- C. Embryonal rhabdomyosarcoma
- D. Teratoma
- E. Neuroblastoma

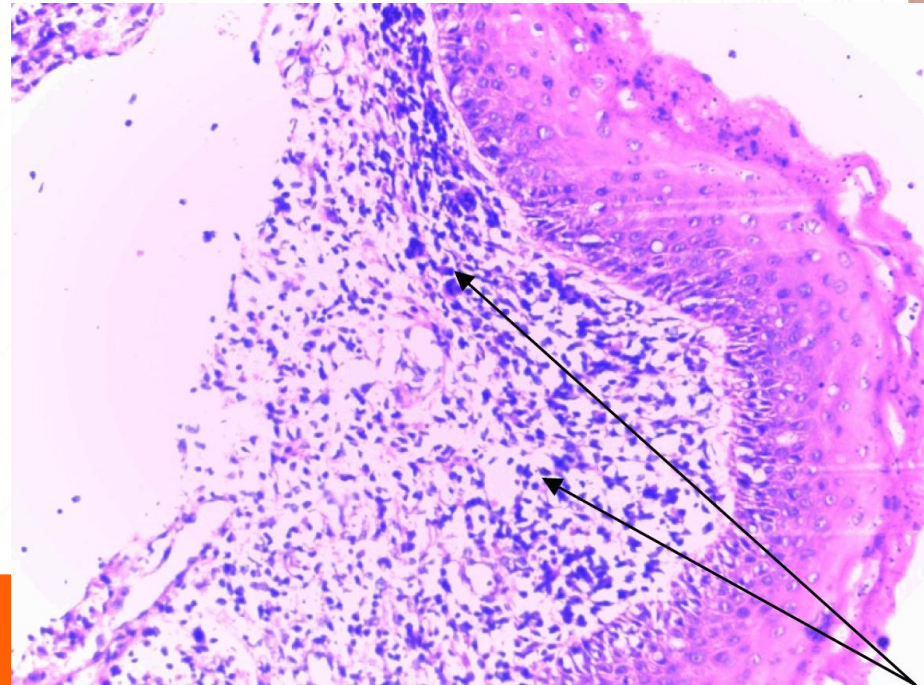
Sarcoma Botryoides or Embryonal Rhabdomyosarcoma

LO: Describe the clinicopathologic features of vaginal embryonal rhabdomyosarcoma (sarcoma botryoides).

- Children < 4yo
- Polypoid, grape-like soft tissue protruding from vagina
- Micro: Cambium layer: dense layer of malignant cells just beneath the vaginal epithelium.
- Cells: Rhabdomyoblasts (primitive skeletal muscle cells) with eosinophilic cytoplasm, sometimes with “strap” or “tadpole” shapes.
- Immunohistochemistry: Desmin (+), Myogenin (+), MyoD1 (+)



<https://mynotes4usmle.tumblr.com/post/43609759641/sarcoma-botryoides-or-embryonal-rhabdomyosarcoma>



Question 11:

A 19-year-old nulligravid woman has a normal pelvic exam.

Where is the squamocolumnar junction (SCJ) most likely located in a healthy adolescent or young adult?

- A. Deep within endocervical canal above external os
- B. Within endocervical canal
- C. On the ectocervix
- D. At the internal os
- E. Not visible

Question 12:

A 58-year-old postmenopausal woman undergoes routine cervical screening. On speculum exam, the squamocolumnar junction is not visible. Cytology reveals atypical squamous cells of undetermined significance (ASC-US).

Which of the following best explains the location of the transformation zone in this patient?

- A. On the ectocervix due to estrogen-induced eversion
- B. Within the endocervical canal due to age-related regression
- C. At the internal os due to pregnancy-related changes
- D. Deep in the cervical stroma due to chronic inflammation
- E. Not present in postmenopausal women

Question 13:

A 44-year-old presents with postcoital bleeding. Workup reveals HSIL and a lesion in the TZ.

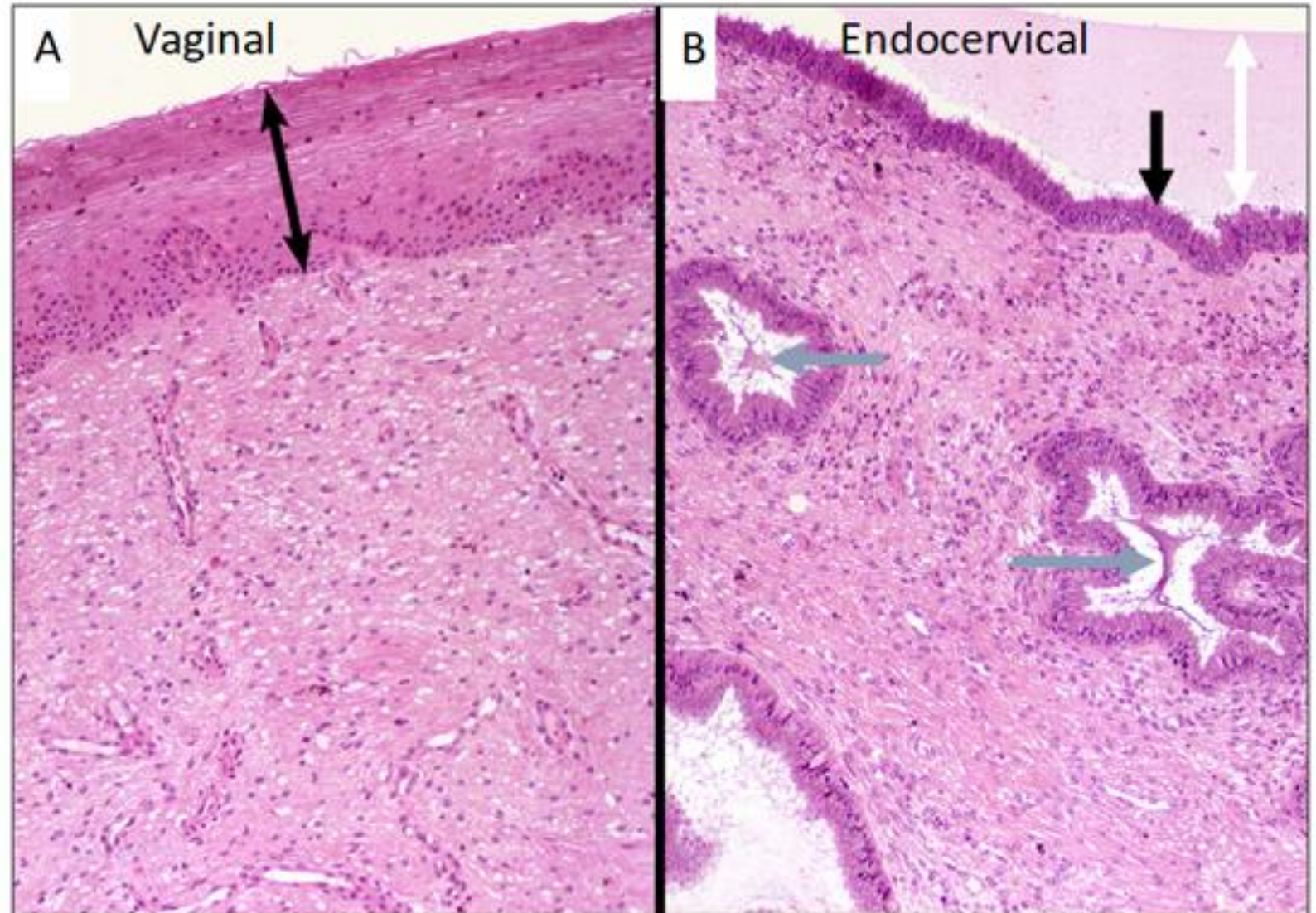
Which functional concept links the TZ's histology to her risk?

- A. Mature glycogenated squamous cells are resistant to HPV
- B. Immature metaplastic squamous cells in the TZ are susceptible to HPV oncogenesis
- C. Columnar epithelium of ectocervix is most resistant to integration
- D. The SCJ location is fixed after puberty
- E. HPV replicates only in basal cells, not maturing cells

Cervical Histology:

LO: Correlate the histological structure of the cervix with its function.

- The ectocervix, like vagina (i.e., the part of the cervix protruding into the vagina) is lined with squamous epithelium
- The epithelial lining of the endocervical canal is columnar mucous-secreting glandular epithelium
- Infolds of this epithelium form plicae palmatae (“palm folds”); these are known as cervical crypts or glands.
- In the presence of high estrogen levels at ovulation, the cells lining the crypts secrete abundant alkaline secretions (grey arrows). This mucus lubricates the vagina and nourishes spermatozoa.



The squamocolumnar junction SCJ and the Transformation Zone

LO: Correlate the histological structure of the cervix with its function.

Predict functional outcomes that result from altered structure and function of the uterine cervix (e.g., cervical cancer).

1. Fetal stage

- The SCJ (where squamous epithelium of the ectocervix meets columnar epithelium of the endocervix) is located relatively low on the cervix.
- During development, the columnar epithelium extends outward.

2. Pre-pubertal stage

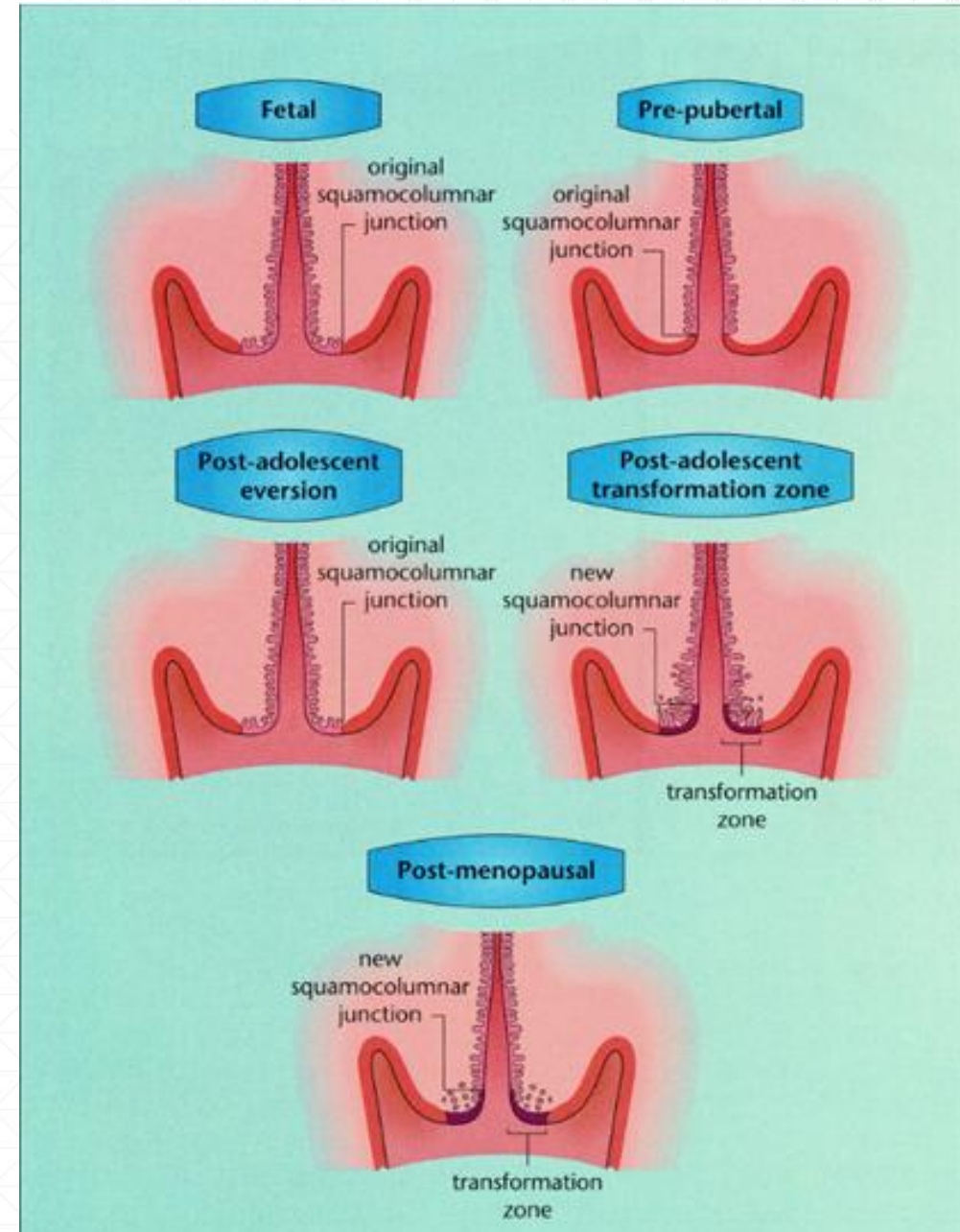
- The cervix is smaller, and the SCJ moves inward (up in the canal) compared to the fetal stage.
- This is because of cervical growth and changes in the hormonal environment (low estrogen state before puberty).

3. Post-adolescent (eversion and transformation zone)

- With puberty and estrogen stimulation, the cervix enlarges, and columnar epithelium everts outward onto the ectocervix.
- This exposed columnar epithelium undergoes squamous metaplasia, forming the transformation zone (high-risk site for HPV infection and cervical neoplasia). **The area between the original SCJ and the newly formed SCJ is known as the transformation Zone (TZ).**

4. Post-menopausal stage

- Due to low estrogen, the cervix shrinks, and the SCJ recedes further upward into the endocervical canal.
- The transformation zone becomes smaller and less accessible.



Question 14:

A 32-year-old woman presents for routine cervical cancer screening. Her Pap smear reveals cells with enlarged nuclei, irregular nuclear contours, and increased nuclear-to-cytoplasmic ratio. Colposcopic biopsy confirms CIN II. She is counseled on management options.

Which of the following best describes the morphologic and clinical features of CIN II?

- A. Koilocytosis with minimal nuclear atypia; spontaneous regression is common
- B. Full-thickness epithelial dysplasia; high risk of progression to invasive carcinoma
- C. Moderate dysplasia involving the lower two-thirds of the epithelium; variable regression or progression
- D. Mild dysplasia confined to the basal third of the epithelium; typically managed with observation
- E. Invasion of basement membrane by atypical cells; requires staging and oncologic treatment

Question 15:

A 49-year-old woman presents with abnormal vaginal bleeding. She has not had a Pap smear in over 10 years. Pelvic examination reveals a friable, exophytic cervical mass. Biopsy shows nests of malignant cells invading the cervical stroma, with keratin pearls and intercellular bridges.

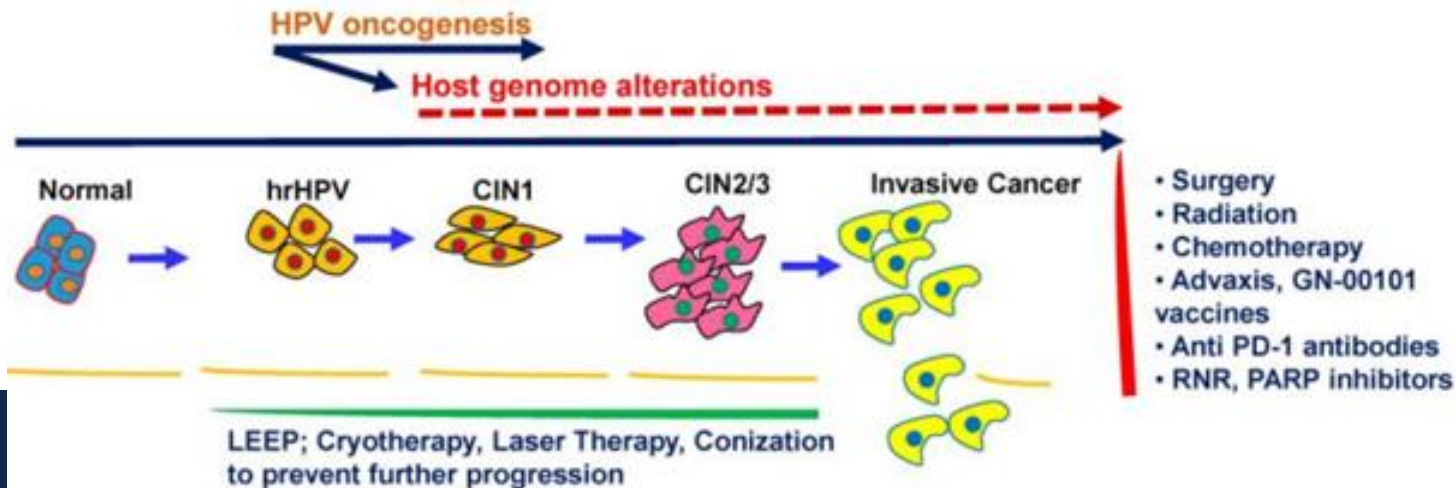
Which of the following best describes the diagnosis and its associated features?

- A. CIN III; full-thickness dysplasia without stromal invasion
- B. Adenocarcinoma in situ; arises from glandular epithelium
- C. Invasive squamous cell carcinoma; associated with high-risk HPV types
- D. CIN I; koilocytosis and mild nuclear atypia
- E. High-grade glandular dysplasia; precursor to squamous carcinoma

Pathogenesis of Cervical Intraepithelial Neoplasia (CINs)

LO: Discuss the common human papillomavirus types that affect the cervix and discuss the pathogenesis of cervical dysplasia and neoplasia, and cervical screening methods and prevention.

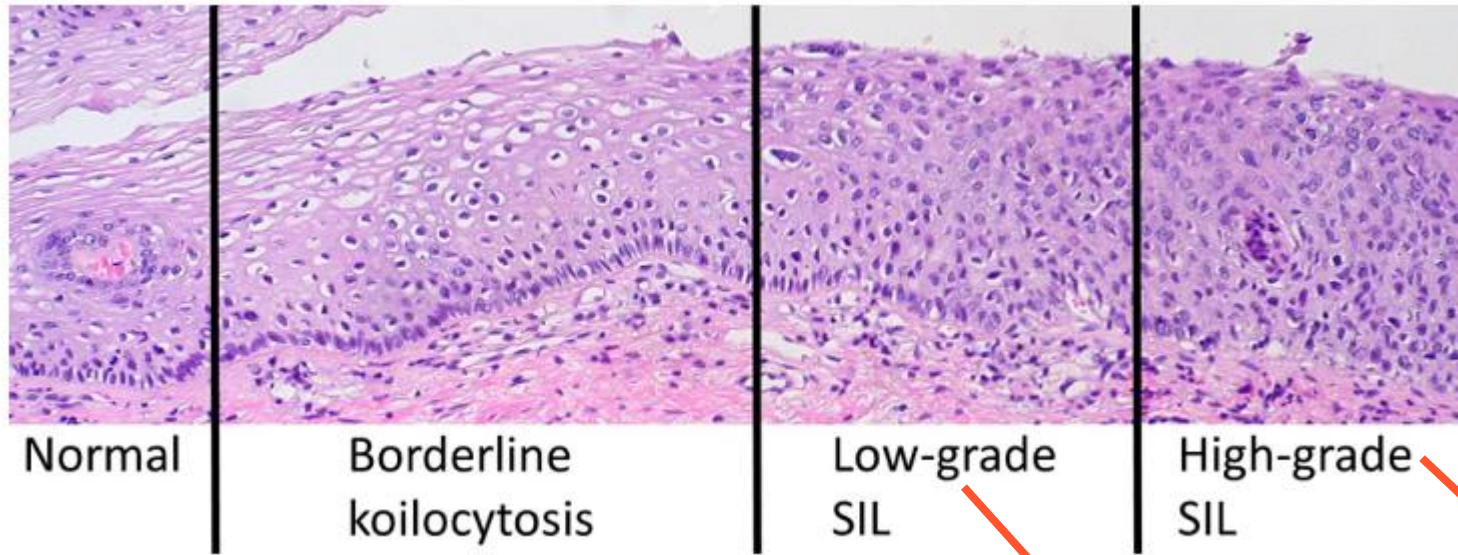
- CINs are the precursor lesions of cervical carcinomas.
- Arise from normal cervical epithelium; low-grade and high-grade CINs.
- The most important factor in the pathogenesis is the High-risk HPV (hrHPV).
- Other contributing factors (smoking, immune status)
- Low oncogenic risk HPV (types 6 and 11) and high oncogenic risk (types 16 and 18; longer to clear) based on their genotypes
- High-risk HPVs are the main pathogenic factors of squamous cell carcinomas of other tissues (e.g., oropharynx, tonsils, anus, vagina, vulva, and penis).
- Risk factors associated with HPV infections are early age at first intercourse, multiple sexual partners, and persistent infection with high-risk strains of HPV.
- The hrHPV infection into the cervical epithelium results in host genome alterations, leading to the silencing of various tumor-suppressor factors on one hand and inducing aberrant functioning of various tumor-promoting factors on the other
- The imbalance and instability caused by various hrHPV-derived oncogenic factors drive neoplastic progression



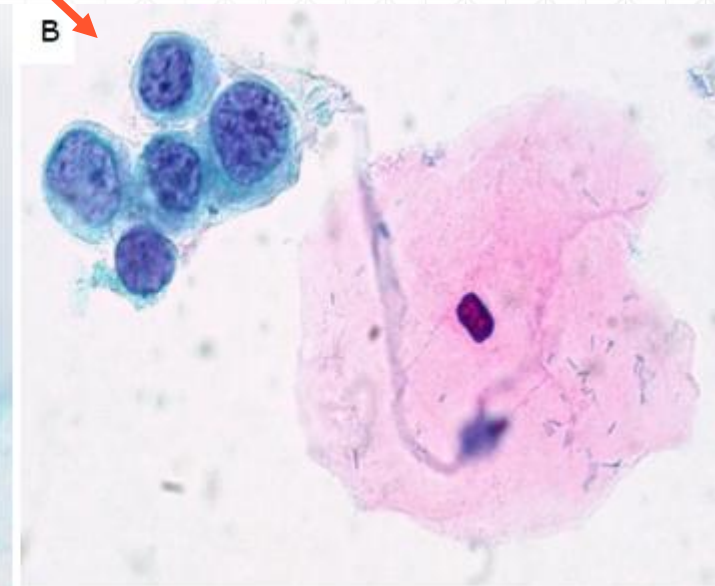
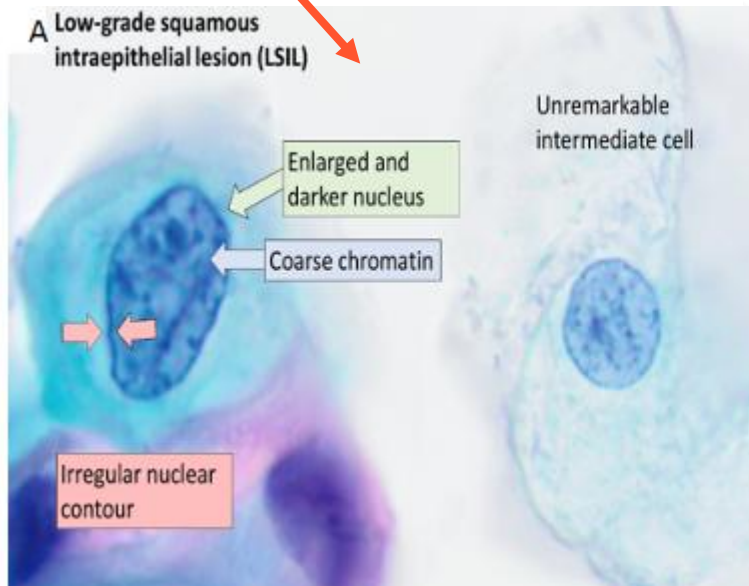
Pathogenesis of Cervical Intraepithelial Neoplasia (CINs)-Cont

- The major oncoproteins are E5, E6, and E7 encoded by the HPV genome
- They disrupt the normal functioning of the major histocompatibility complex I (MHC class I), p53 and Rb, Notch1, Wnt, MAPK, mTOR, and STAT-associated pathways, which are central players controlling normal cellular growth, differentiation, and immune function
- HPVs induce carcinogenesis through dysregulation of the activity of tumor suppressor proteins such as p53 and RB by the viral E6 and E7 proteins, respectively---prevents the normal arrest of TZ cells in the G1 phase of the cell cycle.
- The viral E6 upregulates telomerase and binds to p53, leading to its destruction.
- E7 binds to the RB protein, leading to the release of the E2F transcription factors from the inhibitory effect of RB. This results in progression through the cell cycle.
- High-risk HPVs also integrate into the host cell genome. This integration results in increased expression of the viral E6 and E7 proteins, and host cell genomic instability----acquisition of prooncogenic mutations.
- Low-risk HPVs do not integrate into the host genome.
- The **mechanism for low-grade dysplasia** is **accumulation of viral particles (DNA)**, producing the classic koilocyte
- Although CIN II/III has a higher risk for progression to invasive cervical carcinoma, not all women experience this; in fact, only about 20% progress.
- **VIN (vaginal intraepithelial neoplasia) is similar in appearance to CIN but affects the vagina and affects an older population. It may represent a field effect from viral infection of the cervix.**

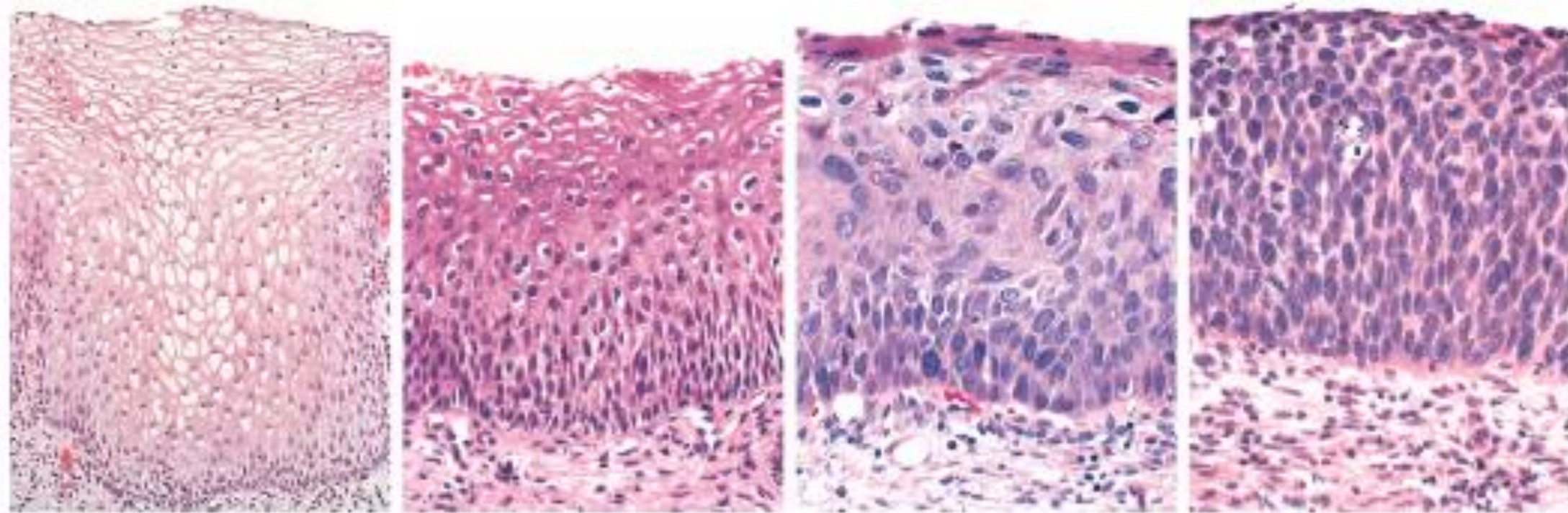
Spectrum from normal to high-grade SIL



There is no perfect correlation (e.g., a person with LSIL (Low-grade squamous intraepithelial lesion) on Pap smear can have CIN II/III on biopsy, and the reverse is true), but in general, there is good correlation.



Spectrum from normal to high-grade SIL- H&E



Healthy

LSIL (CIN I)

HSIL (CIN II)

HSIL (CIN III)

FIG. 17.6

Spectrum of squamous intraepithelial lesions (SILs) with healthy squamous epithelium for comparison: LSIL (CIN I) with koilocytotic atypia; HSIL (CIN II) with progressive atypia in all layers of the epithelium; and HSIL (CIN III) with diffuse atypia and loss of maturation (*far right image*). CIN, Cervical intraepithelial neoplasia; HSIL, high-grade squamous intraepithelial lesion; LSIL, low-grade squamous intraepithelial lesion.

Spectrum from normal to high-grade SIL- Immunostaining

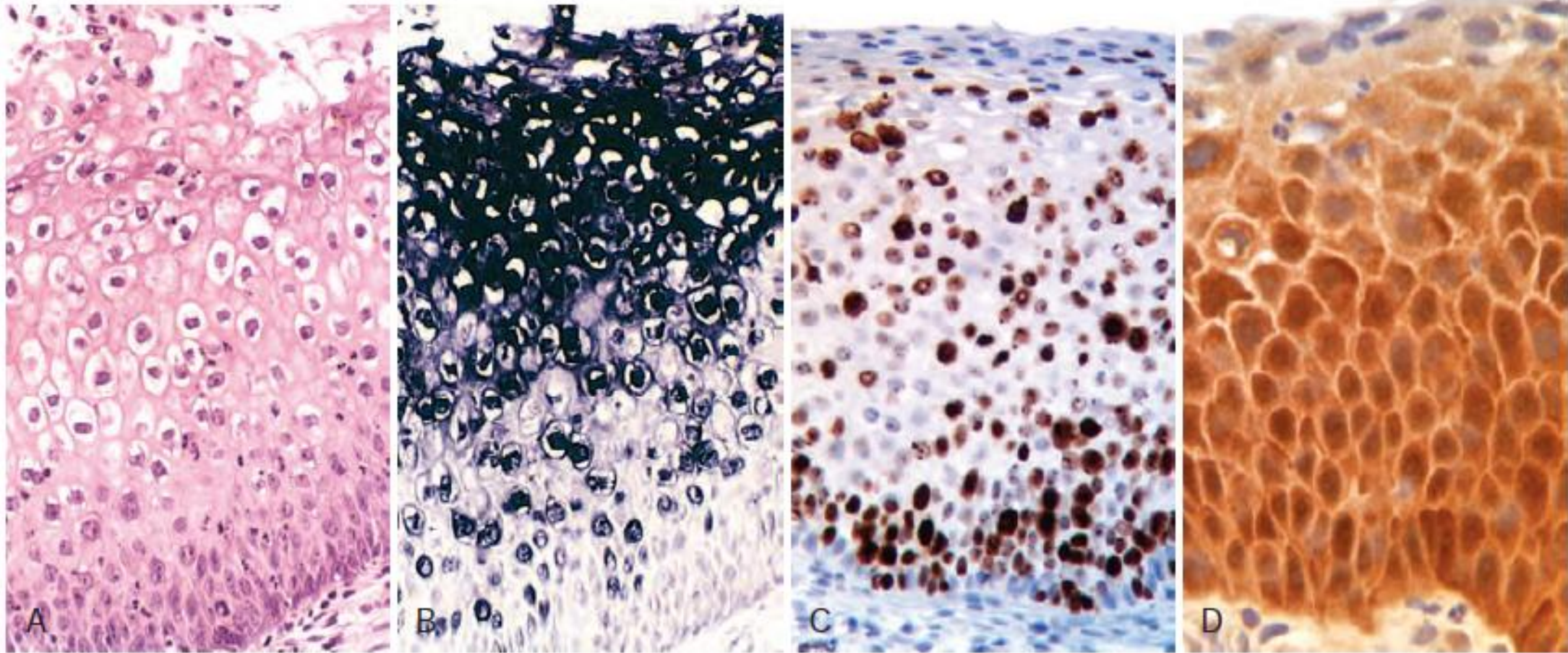
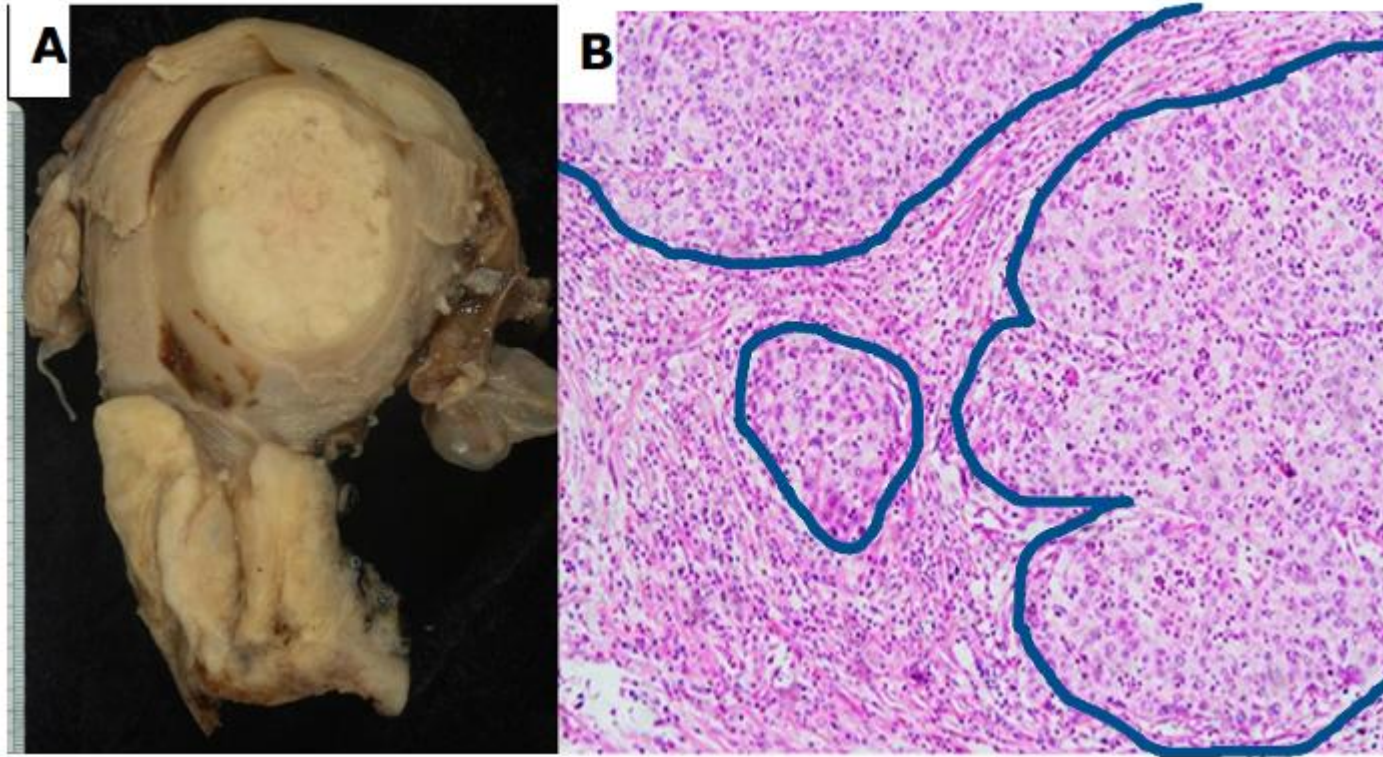


Figure 22.15 (A) Low-grade squamous intraepithelial lesion: routine hematoxylin and eosin staining shows koilocytic change, seen as nuclear enlargement and perinuclear “halos.” (B) In situ hybridization test for human papillomavirus deoxyribonucleic acid (HPV DNA). The dark granular staining denotes HPV DNA, which is most abundant in the superficial koilocytes. (C) Diffuse positivity for the proliferation marker Ki-67 (seen as brown nuclear staining), illustrates expansion of the proliferating cells from the normal basal location to the superficial layers of the epithelium. (D) Up-regulation of the cyclin-dependent kinase inhibitor p16 (seen here as brown staining) characterizes high-risk HPV infections.

Invasive Squamous Cell Carcinoma of the Cervix



LO: Describe the morphologic features of the spectrum of cervical dysplasia and neoplasia derived from both cytologic and tissue specimens and clinical outcomes associated with each.

Invasive Squamous Cell Carcinoma of the Cervix. Large squamous carcinoma (bottom of the picture) has obliterated the cervix and invaded the lower uterine segment (A). The uterus also has a round leiomyoma up higher. Poorly differentiated squamous cell carcinoma of the cervix (B). Nests of malignant cells (blue free lines). Note the stromal desmoplastic reaction between the nests.

Thank You

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